

VNIR SPECTRAL ANALYSES OF LUNAR METEORITES: MINERALOGICAL INSIGHTS AND IMPLICATIONS FOR FUTURE LUNAR IMAGERS. J. Flahaut¹, R. Rossello¹, C. Peignaux¹, M. Martinot¹, J. Cauzid², N. Théret³, C. Virmontois³. ¹Université de Lorraine, CNRS, CRPG, Nancy F-54000, France (Jessica.flahaut@univ-lorraine.fr); ²Université de Lorraine, CNRS, GeoRessources, Nancy F-54000, France; ³Centre National d'Etudes Spatiales (CNES), Toulouse F-31000, France.

Introduction: Visible Near-InfraRed (VNIR) spectroscopy is a key technique to survey the mineralogical composition of distant, rocky objects in the Solar System. The use of VNIR spectroscopy has already led to many discoveries pertaining to the Moon geologic history from orbit onboard the Clementine (NASA, 1994), Smart-1 (ESA, 2003), SELENE (JAXA, 2007) and Chandrayaan-1 (ISRO, 2008) missions (e.g., [1, 2]). VNIR spectroscopy has also been employed at a few occurrences *in-situ* but has yet to be popularized for future lunar missions. Whereas the Chang'E-3 and 4 Yutu rovers carried VNIR point spectrometer as part of their scientific payloads, the JAXA SLIM lander featured a VNIR multispectral camera that emphasized the added value of spectral imagery by identifying olivine-rich rocks at a distance from the spacecraft [3-4].

The present paper investigated VNIR spectral images of 3 lunar meteorites to: 1) infer their mineralogical composition from this non destructive, remotely sensed approach, and 2) use the laboratory data to further test and calibrate two compact VNIR spectral imagers currently developed at CNES for future lunar missions.

Samples description: Three lunar meteorites that were available as slabs in the private collection of researchers at CRPG Nancy were investigated:

- Bechar 006: Feldspathic breccia, found in Algeria in 2022.
- NWA 11212: Feldspathic breccia, purchased in Morocco in 2016.
- NWA 13739: Feldspathic breccia, found in Algeria in 2020.

Data acquisition: The meteorite slices were analyzed in the CRPG hyperspectral remote sensing laboratory (<https://crpg.univ-lorraine.fr/en/hyperspectral-remote-sensing-en/>) with both the Hypspec VNIR 3000N (400-1000 nm) and the Hypspec SWIR 640 (960-2500 nm) spectral cameras. Using microscopic objectives, a spatial resolution of 15 microns/px (VNIR) to 32 microns/px (SWIR) can be achieved. Spectral sampling is 2 nm in the VNIR, 4.38 nm in the SWIR (e.g., [5]).

In addition, the chemistry of the meteorites was investigated with X-ray microfluorescence (μ XRF) measurements carried out at GeoRessources Nancy, France (<https://georessources.univ-lorraine.fr/en/content/micro-xrf>).

Data reduction: The hyperspectral data cubes were converted to reflectance using the Hypspec software and a calibrated panel of known reflectance. μ XRF data cubes were converted into elemental images using spectral fitting in each pixel with PyMCA [6].

In order to mimic the signal of the two CNES multispectral imagers, an IDL/ENVI routine was coded to match the data to the spectral resolution of the future VNIR imagers. Two VNIR imagers have been recently developed at CNES, building on the compact CASPEX camera system and adding customized SILIOS filters (CAM-8) or IMEC filters (CAM-25) [7]. CAM-8 is a VNIR spectral imager that operates with 8 bands between 550 and 950 nm with 4x4 multispectral spatial pattern, using Full Width at Half Maximum (FWHM) between 20-30 nm to maximize flux. CAM-25 should cover the range 660 – 960 nm with a 5x5 multispectral pattern (hence, 25 bands) and narrow FWHM (6-18 nm) to maximize spectral resolution. To be as close as possible to the realistic signal of the instrument, the transmission and quantum efficiency curves of the filters and optics were used to simulate the resulting spectra.

Results: Chemistry and laboratory spectra.

Bechar 006 is a fine-grained polymict breccia dominated by plagioclase, pyroxene, olivine and a crystallized melt matrix (Figure 1a,b). μ XRF data confirms the presence of accessory minerals such as ilmenite, troilite, and Ni-bearing metal. Terrestrial alteration on the outer crust and within fractures is evidenced by high Ba and Ca concentrations and spectral signatures of calcite, barite and Al-phyllsilicates in the SWIR (Figure 1d, e). SWIR spectra further allow us to infer the presence of pigeonite (spectral features centered ~ 950 and 1970 nm) and augite (spectral features centered ~ 1015 and 2180 nm) as well as anorthite with a weak 1300 nm centered absorption. Plagioclase are however commonly altered as evidenced by the presence of narrow OH/H₂O absorptions ~ 1420 and 1920 nm respectively. Spinel is also detected by a broad absorption after 1380 nm and confirmed by elevated Cr, Fe and Mg concentration in the μ XRF maps. In the VNIR domain (Figure 1c), olivine can be easily identified from its absorption band at 630 nm and a diagnostic drop after 680 nm. Pigeonite is characterized by a reflectance peak around 750 nm, followed

by a strong decrease in reflectance towards higher wavelengths. Spinel has a very low average reflectance level (<0.1) and a weak positive trend in the 550-950 nm range. Plagioclase are however difficult to identify from their VNIR features, which could be partly due to their alteration.

NWA 11212 is a breccia composed of small angular mineral grains. Spectrally, it is dominated by pyroxene (pigeonite) and featureless spectra, but olivine and terrestrially altered plagioclase spectra are still observable.

NWA 13739 is a complex breccia composed of both feldspathic and gabbroic clasts, and dominated by signatures of pyroxenes (pigeonite and augite), olivine, plagioclase and featureless spectra in both the VNIR and SWIR ranges.

CAM-8 spectra simulations. Figure 1b displays a color composite of Bechar 006 as seen with CAM-8 bands B1 (550 nm), B3 (652 nm) and B8 (940 nm). This simple RGB composite already highlights the spectral diversity of the sample, with strong pyroxene signatures mapped in light green (pigeonite) to turquoise (augite), olivine mapped in darker, olive green, terrestrial alteration mapped in light purple/blue and plagioclase / glass signals mapped as dark blue. Associated simulated spectra are presented in figure 1c (dashed lines). Although mineral identification was initially done with the help of both VNIR and SWIR spectra, most mineral features are still recognizable from the VNIR absorptions only. Low calcium pyroxene and pigeonite show a peak in reflectance at CAM-8 band 5 (769 nm), whereas the peak can be shifted to band 6 (832 nm) or 7 (880 nm) for high calcium pyroxene. Pure olivine shows a strong decreasing slope from CAM-8 band 3 (652 nm) onwards. Plagioclase and oxides (including spinel) should be discernable by their relatively high and low reflectance values, respectively. The three meteorites yielded similar results. Distinct trends could be further highlighted with selected band rati-

os. Importantly, simulating the camera response taking into account the filters sensitivity results in spectra that preserve key spectral features.

Discussion and perspectives: This study demonstrates the potential of VNIR imagery for quick mineral identification. Additional petrologic analyses will be carried out to confirm the mineral detections. The method is remote and non destructive, making it an ideal tool for in situ characterization as well as preliminary laboratory analyses of rare material. We further demonstrated that the use of as little as 8 filters in a restricted range of the VNIR domain should be sufficient to identify most of the main rock-forming minerals at the lunar surface, taking into account the instrument sensitivity. Ongoing work include: 1) the development of customized RGB and spectral parameters that could be used as quickview tools for both CAM-8 and CAM-25; 2) adaptation of our routine to contribute to the development of a SWIR multispectral camera with customized filters; 3) “demoisacing” activities (e.g., interpolation) at CNES, to reproduce a full resolution image from the multispectral spatial pattern.

References: [1] Shkuratov Y. et al. (2025), *Elsevier*, ISBN 9780128179727. [2] Mangold N. et al. (2019), *Oxford Research Encyclopedia of Planetary Science* [3] Ohtake M. et al. (2025), *LPSC LXI*, 1912. [4] . Nakauchi Y. et al. (2025), *LPSC LXI*, 1924. [5] Barthez M. et al. (2023), *JGR*, 128(8), e2022JE007680. [6] (Solé V. A. et al. (2007), *Spec. Acta Part B*, 62(1), 63-68. [7] Virmondois C. et al. (2025), *LPSC LXI*, 1912.

Figure 1: Spectral analysis of lunar meteorite Bechar 006. a) Default laboratory RGB (637, 544, 460nm). b) CAM-8 color composite of bands 1, 3, 8. c) Selected VNIR spectra (full lines) collected on the laboratory VNIR hyperspectral cube (2x2 avg) versus simulated VNIR spectra (dashed lines) using the instrument spec. d) Selected spectra collected on the laboratory SWIR hyperspectral cube (2x2 avg). e) USGS reference spectra of selected terrestrial minerals are shown for comparison.

